make_sjsdm_structured_regularization_candidates.R

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`make_sjsdm_structured_regularization_candidates`
Make Structured sjSDM Regularization Candidates

Description

Builds a deterministic first-pass coordinate search around one reference regularization candidate without creating a Cartesian product.

Usage

```
make_sjsdm_structured_regularization_candidates(
  alpha_cov = 0.5,
  alpha_coef = 0.5,
  alpha_spatial = 0.5,
  lambda_cov_reference = 0.1,
  lambda_coef_reference = 0.1,
  lambda_spatial_reference = 0.1,
  lambda_values = base::c(0, 0.01, 0.03, 0.1, 0.3, 1)
)
```

Arguments

`alpha_cov`, `alpha_coef`, `alpha_spatial`
: Fixed finite elastic-net mixing values between zero and one.

`lambda_cov_reference`, `lambda_coef_reference`
: Reference covariance and abiotic-coefficient lambda values.

`lambda_spatial_reference`
: Reference spatial-coefficient lambda value.

`lambda_values`
: Unique finite non-negative values searched separately along each lambda axis. The vector must contain every reference value, with each reference strictly inside the search range.

Details

The reference candidate appears once. Every other candidate changes exactly one lambda while holding the remaining lambdas at their reference values. With six shared lambda values and one shared reference, this produces 16 candidates instead of a 216-row Cartesian product. Input order does not affect candidate identifiers.

Value

Named list containing 'data_candidates', which follows the production tuning-candidate schema, and 'data_search_design', which records the varied axis, axis value, reference status, and lower/upper boundary flags.

Examples

```
make_sjsdm_structured_regularization_candidates(
  lambda_values = c(0, 0.01, 0.03, 0.1, 0.3, 1)
)
```

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